

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 208
Course Title : Database Systems Lab
Experiment No. : 04

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Intake: 53
Section: 01
Program: B.Sc. Engg. in CSE

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Date of Submission:

Signature of Teacher

1. Perform Cartesian Product Operation between these two relation.

Select *from Instructor,teaches;

Id	Name	dept_name	salary	Id	c_id	section_id
10101	Srinivasan	CSE	65000	10101	CSE_101	1
10101	Srinivasan	CSE	65000	12121	CSE_111	2
10101	Srinivasan	CSE	65000	13131	CSE_311	3
12121	Wu	FIN	90000	10101	CSE_101	1
12121	Wu	FIN	90000	12121	CSE_111	2
12121	Wu	FIN	90000	13131	CSE_311	3
15151	Mozart	Music	40000	10101	CSE_101	1
15151	Mozart	Music	40000	12121	CSE_111	2
15151	Mozart	Music	40000	13131	CSE_311	3
22222	Einstein	Physics	90000	10101	CSE_101	1
22222	Einstein	Physics	90000	12121	CSE_111	2
22222	Einstein	Physics	90000	13131	CSE_311	3
32343	Said	History	60000	10101	CSE_101	1
32343	Said	History	60000	12121	CSE_111	2
32343	Said	History	60000	13131	CSE_311	3
33456	Gold	Physics	87000	10101	CSE_101	1
33456	Gold	Physics	87000	12121	CSE_111	2
33456	Gold	Physics	87000	13131	CSE_311	3
45565	katz	CSE	75000	10101	CSE_101	1
45565	katz	CSE	75000	12121	CSE_111	2
45565	katz	CSE	75000	13131	CSE_311	3
58583	cali	History	62000	10101	CSE_101	1
58583	cali	History	62000	12121	CSE_111	2
58583	cali	History	62000	13131	CSE_311	3
76543	Singh	FIN	80000	10101	CSE_101	1

2. Find those instructors who teaches any of the courses.

Select name, C_id from instructor INNER JOIN teaches where
Instructor.ID=teaches. ID;

✓ Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
SELECT Name, C_Id FROM instructor INNER JOIN teaches WHERE instructor.Id = teaches. Id;
```

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☐ Show all

Number of rows:

25

Filter rows:

Search this table

Extra options

Name	C_Id
Srinivasan	CSE_101
Wu	CSE_111

3. Find only instructor names and course id for instructors in the Computer Science department.

Select name, C_id from instructor INNER JOIN teaches where Instructor.ID=teaches. ID AND dept_name= "CSE";

✓ Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

```
SELECT Name, C_Id FROM instructor INNER JOIN teaches WHERE instructor.Id = teachers. Id AND dept_name = "CSE";
```

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☐ Show all | Number of rows: 25 | Filter rows:

Extra options

Name	C_Id
Srinivasan	CSE_101

4. Find the total no. of tuples in "Instructor" relation.

Select count(*) from Instructor.

✓ Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)

```
Select count(*) from Instructor;
```

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Extra options

count(*)
12

5. Answer Q2 using Natural Join.

Select Name, C_Id from instructor NATURAL JOIN teachers;

✓ Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
Select Name, C_Id from instructor NATURAL JOIN teachers;
```

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Extra options

Name	C_Id
Srinivasan	CSE_101
Wu	CSE_111

6. Answer Q3 using Natural Join.

Select Name, C_Id from instructor NATURAL JOIN teachers where dept_Name='CSE';

✓ Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

```
Select Name, C_Id from instructor NATURAL JOIN teachers where dept_Name= 'CSE';
```

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☐ Show all | Number of rows: 25 ▼ Filter rows:

Extra options

Name	C_Id
Srinivasan	CSE_101

7. Perform Left Outer Join.

SELECT * FROM `the account relation` NATURAL LEFT OUTER JOIN `the depositor relation`;

✓ Showing rows 0 - 6 (7 total, Query took 0.0002 seconds.)

```
SELECT * FROM `the account relation` NATURAL LEFT OUTER JOIN `the depositor relation`;
```

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☐ Show all | Number of rows: 25 ▼ Filter rows:

Extra options

account_number	branch_name	balance	customer_name
A-102	Perryridge	400	Hayes
A-101	Downtown	500	Johnson
A-201	Brighton	900	Johnson
A-217	Brighton	750	Jones
A-222	Redwood	700	Lindsay
A-215	Mianus	700	Smith
A-305	Round Hill	350	Turner

8. Perform Right Outer Join.

SELECT * FROM `the account relation` NATURAL RIGHT OUTER JOIN `the depositor relation`;

Showing rows 0 - 6 (7 total, Query took 0.0002 seconds.)

```
SELECT * FROM `the account relation` NATURAL RIGHT OUTER JOIN `the depositor relation`;
```

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☐ Show all | Number of rows: 25 | Filter rows:

Extra options

account_number	customer_name	branch_name	balance
A-101	Johnson	Downtown	500
A-102	Hayes	Perryridge	400
A-201	Johnson	Brighton	900
A-215	Smith	Mianus	700
A-217	Jones	Brighton	750
A-222	Lindsay	Redwood	700
A-305	Turner	Round Hill	350

9. Perform Full Outer Join.

(SELECT * FROM `the account relation` NATURAL LEFT OUTER JOIN `the depositor relation`;) UNION (SELECT * FROM `the account relation` NATURAL RIGHT OUTER JOIN `the depositor relation`);

Showing rows 0 - 13 (14 total, Query took 0.0007 seconds.)

```
(SELECT * FROM `the account relation` NATURAL LEFT OUTER JOIN `the depositor relation`) UNION (SELECT * FROM `the account relation` NATURAL RIGHT OUTER JOIN `the depositor relation`);
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

account_number	branch_name	balance	customer_name
A-102	Perryridge	400	Hayes
A-101	Downtown	500	Johnson
A-201	Brighton	900	Johnson
A-217	Brighton	750	Jones
A-222	Redwood	700	Lindsay
A-215	Mianus	700	Smith
A-305	Round Hill	350	Turner
A-101	Johnson	Downtown	500
A-102	Hayes	Perryridge	400
A-201	Johnson	Brighton	900
A-215	Smith	Mianus	700
A-217	Jones	Brighton	750
A-222	Lindsay	Redwood	700
A-305	Turner	Round Hill	350